# Cecilia Sgalletta

*Curriculum Vitae*

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## Academic experience

Nov 2025 – Oct 2027 **Humboldt fellowship**, *Institut für Theoretische Astrophysik*, Heidelberg, Germany
I am currently an Humboldt fellow working on the project titled "The challenge of binary black hole models: why is the predicted merger rate too high?".

Oct 2021 – Sep 2025 **PhD in *Astrophysics and cosmology***, *SISSA*, Trieste, Italy
Supervisor: Prof. Mario Spera
Co-supervisors: Prof. Andrea Lapi and Prof. Michela Mapelli
Throughout my PhD, I've worked on several projects related to the study of binary evolution processes and the formation of binary compact object systems, with a special focus on binary neutron star systems ( Sgalletta et al. 2023), merging binary black holes (Sgalletta et al. 2025) and the physics of common envelope (Sgalletta et al. 2026, submitted to A&A). My PhD thesis can be read at this link.

Nov 2023 – Jan 2024 **Scientific collaboration**, *Institut für Theoretische Astrophysik*, Heidelberg, Germany
Collaboration with Prof. Michela Mapelli
During my visit I investigated the merger rate density of binary black holes across cosmic time, which led to Sgalletta et al. 2025.

## Education

2019 – 2021 **Master's degree in *Physics of the fundamental interactions***, *University of Padova*, Padova, Italy, 110/110 with honors
Supervisor: Prof. Michela Mapelli
Co-supervisors: Dr. Maria Celeste Artale and Dr. Giuliano Iorio.

Summer 2020 **Summer internship**, *University of Padova*, Padova, Italy
Supervisor: Prof. Michela Mapelli

2016 – 2019 **Bachelor's degree in *Physics***, *University of Padova*, Padova, Italy, 104/110
Supervisor: Prof. Sabino Matarrese
Co-supervisor: Dr. Daniele Bertacca

## Talks and summer schools

I presented my research activities in several conferences, among which **three invited talks and a review talk**. Slides of some presentations can be found at this link.

09 – 13 Mar 2026 **ABETE: Avalanche of Bright sources towards the ET Era. The near-future prospects and challenges of multi-messenger astronomy**, Sexten, Italy
Invited talk.

06 Feb 2025 **Invited seminar**, *BICOCCA University*, Milan, Italy
Invited seminar "Binary black hole merger rates: the role of the metal-dependent star formation rate", based on my recent work Sgalletta et al. 2025.

29 Sep – 04 Oct 2024 **Riding the waves into the future, TUTT2024**, Valona, Albania
Invited review talk "Population models and rates" on what we already know about the population of gravitational wave sources and the present challenges in binary population synthesis codes.

15 – 19 Jul 2024 **41st Liège International Astrophysical Colloquium: The eventful life of massive star multiples**, Liège, Belgium
Talk on the "Binary neutron star populations in the Milky Way ".

10 – 14 Jun 2024 **360º Approach to Common Envelope Evolution**, Barcelona, Spain
Workshop by invitation only, centered on the unsolved problems of the common envelope evolution. Presentation of the poster "Challenges for binary population synthesis: towards a better modelling of the common envelope".

29 Apr – 03 May 2024 **Gravitational waves: a new ear on the chemistry of galaxies**, Leiden, The Netherlands
Workshop at the intersection of gravitational-wave astrophysics and galactic chemical evolution.

04 – 06 Dec 2023 **Forging a New Synthesis Between Supernova Theory and Observation**, Princeton
Workshop by invitation only, on the supernova explosion physics.

18 Oct 2023 **Invited seminar**, *University of Padova*, Padova, Italy
Invited seminar on "Modeling the population of Galactic binary neutron stars", focused on the pivotal role of binary neutron stars in modern astrophysics and their multimessenger nature.

04 – 06 Oct 2023 **The Fourth Gravi-Gamma-Nu Workshop**, *GSSI*, L'Aquila, Italy
Talk about "Binary Neutron Star populations in the Milky Way"

10 – 14 Jul 2023 **Gravitational-wave populations: what's next?**, *BICOCCA University*, Milano, Italy
Short talk about "Binary Neutron Star populations in the Milky Way"

05 – 09 Sep 2022 **Amaldi Research Center Summer School**, Paestum, Italy
Summer school on gravitational physics and astrophysics

25 – 29 Jul 2022 **The ONASSIS foundation science lecture series 2022 in physics**, Heraklion, Crete
Fully funded summer school on gravitational wave theory, analysis and detectors.

21 – 22 Jul 2022 **TEONGRAV annual meeting**, Florence, Italy
Invited talk about "Modeling the population of Galactic binary neutron stars"

06 – 09 Apr 2022 **Spring workshop of physics of data**, *University of Padova*, Venice, Italy
Invited talk about "Pulsars spin-up: a semi-analytical approach"

## Awards and Grants

Nov 2025 – Oct 2027 **Humboldt fellowship**, *Institut für Theoretische Astrophysik*, Heidelberg, Germany

Sep – Dec 2024 **Erasmus+ programme**, *Institut für Theoretische Astrophysik*, Heidelberg, Germany

Nov 2023 – Jan 2024 **Erasmus+ programme**, *Institut für Theoretische Astrophysik*, Heidelberg, Germany

## Students supervision

- Co-supervisor of Till Gero Friedhelm Krause's bachelor thesis (2025), University of Heidelberg.

## Referee experience

**Referee for the Astrophysical Journal**

**Referee for the Monthly Notices of Royal Astronomical Society**

## Memberships

2022 – now **Member of the Einstein Telescope (ET) Research Unit in Heidelberg**

- ET is the European next-generation ground-based gravitational-wave detector.
- I will study in particular the formation and evolution of binary compact object systems as possible gravitational-wave sources.

2022 – now **Member of the LISA collaboration**

- LISA is a space-based gravitational wave observatory.

2022 – 2025 **Member of the TEONGRAV Trieste group**

- TEONGRAV is a national INFN project, with the purpose of modelling gravitational wave sources, such as binary systems of neutron stars and black holes.

2021 – now **Member of the Observation Science Board of the Einstein Telescope**

- The Observation Science Board's goal is to identify the main scientific cases for the ET, to write the first Blue Book of ET and to organize a broad scientific community for the exploitation of ET.

## Outreach

Apr 23, 2026 **Girl's day 2026**

- The *girl's day* is a yearly initiative organized by the University of Heidelberg to inspire young generations.
- I coordinated the efforts to organize a workshop for the *girl's day*. The young girls got a glimpse into the physics regulating life and death of black holes. They had the opportunity to gain first-hand contact with the world of research.

2025 **SISSA for schools**

- SISSA for schools is a initiative aimed at students from primary to high school, introducing them to research environments through lectures, first-hand conversations with researchers, and other interactive activities.

Sep 23 – 25 2022 **TriesteNEXT**

- TriesteNEXT is a festival aimed at promoting innovation and scientific research, held every year in Trieste
- I actively participated at the event by organizing an astrophysics-themed crossword for children and writing an outreach story titled "Alla ricerca di Gravità" ("In search of Gravity").

06 – 09 Apr 2022 **Spring workshop of physics of data**, *University of Padova*, Venice, Italy

I gave a talk to the students in physics of data of the University of Padova, with a focus on the computational challenges of my work on pulars' evolution.

## IT knowledge and code development

### Programming languages

- Python (excellent), C++ (good), sh/bash (good), HTML (basic).

### Operating systems

- Linux (very good), Windows (good)

### Codes, developer

- **SEVN** ("Stellar EVolution for N-body", population synthesis code for N-body simulations, Spera et al. 2015, Spera et al. 2017, Iorio et al. 2023): written in C++, SEVN allows to integrate stellar evolution and binary evolution processes (tides, Roche lobe overflow, wind accretion, gravitational wave decay, etc) through interpolation algorithms. SEVN is used mostly to investigate the formation of binary compact objects in the field and in star clusters. Open source code downloadable from this link.

## List of publications

Below are shown my publications. Publications where I am the first author are marked with an asterisk.


[1] * **Cecilia Sgalletta**, Giuliano Iorio, Michela Mapelli, Maria C. Artale, Lumen Boco, Debatri Chattopadhyay, Andrea Lapi, Andrea Possenti, Stefano Rinaldi and Mario Spera. *Binary neutron star populations in the Milky Way.* 526(2):2210–2229, Dec. 2023. DOI: 10.1093/mnras/stad2768.

[2] * **Cecilia Sgalletta**, Michela Mapelli, Lumen Boco, Filippo Santoliquido, Maria C. Artale, Giuliano Iorio, Andrea Lapi and Mario Spera. *The more accurately we model the metal-dependent star formation rate, the larger the predicted excess of binary black hole mergers.* A&A 698, A144 (2025); doi:10.1051/0004-6361/202452757.

[3] * **Cecilia Sgalletta**, Guglielmo Costa, Giuliano Iorio, Kendall Shepherd, Francesco Addari, Alessandro A. Trani, Michela Mapelli, Ugo N. Di Carlo, Andrea Lapi, Alessandro Bressan, Mario Spera. *The impact of envelope binding energies on the merger rate density of binary compact objects.* Submitted to A&A.

[4] Rosalba Perna, Maria C. Artale, Yi-Han Wang, Michela Mapelli, Davide Lazzati, **Cecilia Sgalletta**, Filippo Santoliquido. *Host galaxies and electromagnetic counterparts to binary neutron star mergers across the cosmic time: detectability of GW170817-like events.* 512(2):2654–2668, May 2022. DOI: 10.1093/mnras/stac685.

[5] Giuliano Iorio, Michela Mapelli, Guglielmo Costa, Mario Spera, Gaston J. Escobar, **Cecilia Sgalletta**, Alessandro A. Trani, Erika Korb, Filippo Santoliquido, Marco Dall'Amico, Nicola Gaspari and Alessandro Bressan. *Compact object mergers: exploring uncertainties from stellar and binary evolution with SEVN.* 524(1):426–470, Sept. 2023. DOI:10.1093/mnras/stad1630.

[6] Giuliano Iorio, Stefano Torniamenti, Michela Mapelli, Marco Dall'Amico, Alessandro A. Trani, Sara Rastello, **Cecilia Sgalletta**, Stefano Rinaldi, Guglielmo Costa, B. A. Dahl-Lahtinen, Gaston J. Escobar, Erika Korb, Maria P. Vaccaro, Elena Lacchin, Benedetta Mestichelli, Ugo N. Di Carlo, Mario Spera and Manuel Arca Sedda. *The boring history of Gaia BH3 from isolated binary evolution.* 690:A144, Oct. 2024. DOI: 10.1051/0004-6361/202450531.

[7] Francesco Gabrielli, Andrea Lapi, Lumen Boco, Cristiano Ugolini, Guglielmo Costa, **Cecilia Sgalletta**, Kendall Shepherd, Ugo N. Di Carlo, Alessandro Bressan, Marco Limongi and Mario Spera. *The cosmic rate of pair-instability supernovae.* 534(1):151–172, Oct. 2024. DOI:10.1093/mnras/stae2048.

[8] Marco Dall'amico, Michela Mapelli, Giuliano Iorio, Guglielmo Costa, Stéphane Charlot, Erika Korb, **Cecilia Sgalletta**. *Impact of accretion-induced chemically homogeneous evolution on stellar and compact binary populations.*695:A221, Mar. 2025. DOI:10.1051/0004-6361/202453543.

[9] Filippo Simonato, Stefano Torniamenti, Michela Mapelli, Giuliano Iorio, Lumen Boco, Franca De-Domenico Langer, **Cecilia Sgalletta**. *Impact of stellar winds on the pair-instability supernova rate*. 703:A215, Nov. 2025. DOI:10.1051/0004-6361/202555490.

[10] Kendall Shepherd, Guglielmo Costa, Cristiano Ugolini, Guglielmo Volpato, Diego Bossini, **Cecilia Sgalletta**, Francesco Addari, Alessandro Bressan, Leo Girardi, Mario Spera. *Enhanced Mass Loss of Very Massive Stars: Impact on the Evolution, Binary Processes, and Remnant Mass Spectrum*. 701:A126, Sep. 2025. DOI:10.1051/0004-6361/202555467.

[11] Michela Mapelli, **Cecilia Sgalletta**, Johanna Müller-Horn, Giuliano Iorio, Stefano Rinaldi, Christian Burt, Daniel Marin Pina, Amedeo Romagnolo. *The role of accretion efficiency, natal kicks, and angular momentum transport in the formation of the Gaia black holes*. Apr. 2026, Arxiv e-prints: 2604.12839. DOI: 10.48550/arXiv.2604.12839.

[12] Victor Lopez Oller, Boyuan Liu, Michela Mapelli, Stefano Rinaldi, **Cecilia Sgalletta**, Julia Bodensteiner, Giuliano Iorio, Rebekka Schupp. *Population synthesis of Be X-ray binaries in the Small Magellanic Cloud: angular momentum recycling and stable mass transfer*. Apr. 2026, Arxiv e-prints:2604.26693. DOI: 10.48550/arXiv.2604.26693.